Linfang Zheng

Postdoctoral Fellow · Robotics and Embodied AI

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Profile

My research in robotics and embodied AI develops data-efficient robot learning methods that combine scalable visual experience with limited robot supervision, bridging video understanding, geometric perception, and robot manipulation. I integrate structured perception-to-control interfaces with policy learning toward reliable real-world manipulation. With a background spanning embedded hardware and microelectronics, 3D perception, and manipulation algorithms, I bring a full-stack view of robot systems from sensing to control, and I lead collaborative projects mentoring PhD, master's, and undergraduate researchers.

Employment

The University of Hong Kong (HKU) — Hong Kong, China
Postdoctoral Fellow — Jan. 2025 – Present

- Conduct research in Prof. Jia Pan's group on data-efficient robot learning, vision-based manipulation, and 3D perception.

Southern University of Science and Technology (SUSTech) — Shenzhen, China
Research Assistant — Mar. 2019 – Jan. 2020

DJI Co. Ltd. — Shenzhen, China
Embedded Hardware Engineer — Jul. 2017 – Mar. 2019

Education

University of Birmingham (UoB) — Birmingham, UK
Ph.D. in Computer Science — Jan. 2020 – Oct. 2024

- Thesis title: Visual 6D Object Pose Estimation and Tracking
- Supervisors: Prof. Hyung Jin Chang, Prof. Aleš Leonardis, and Prof. Wei Zhang.

Harbin Institute of Technology — Harbin, China
M.Sc. in Integrated Circuit Engineering — Sep. 2015 – Jul. 2017

- Dissertation title: Design and Implementation of a High-Precision Multi-Channel Lidar Measurement and Control System
- Outstanding Master's Graduate; Silver Award for Outstanding Master's Thesis.

Harbin Institute of Technology — Harbin, China
B.Sc. in Electronic Information Science and Technology — Sep. 2011 – Jul. 2015

Research Agenda & Leadership

Data-Efficient Robot Learning from Visual Experience HKU
Action-free visual experience, structured interfaces, and policy improvement 2024 – Present

- Learning manipulation from action-free video, task-agnostic exploration, and few demonstrations, reducing reliance on robot action labels [2], [4].
- Structured perception-to-control interfaces that connect visual learning to robot control through scene representations, visual foresight, 3D motion tracks, and inverse-dynamics grounding [2], [4].
- Policy improvement and adaptation through segment-level credit assignment for RL post-training and robust language conditioning [3].
- **Ongoing:** dense reward models for reinforcement learning in manipulation; video-guided manipulation without task-specific robot labels; and sim-to-real transfer with a dedicated evaluation benchmark.
- Mentor 12 students (2 PhD, 5 master's, and 5 undergraduate) through collaborative research projects. Mentored projects have led to 3 published papers [2]–[4] and 5 manuscripts under review.

Geometry-Aware Perception for Reliable Robot Manipulation UoB & HKU
Spatial understanding from sensing to robot control 2020 – Present

- 6D object pose estimation, tracking, and category-level refinement under shape variation [6], [8], [10], [11], [P1].
- Depth recovery for non-Lambertian objects, plane extraction, and 3D hand-object understanding for manipulation [1], [5], [7].
- **Ongoing:** task-relevant spatial scene understanding for world action models.

Peer-Reviewed Publications

[1] Zhiming Chen*, **Linfang Zheng***, Kun Zhang, Hyung Jin Chang, Wei Zhang, Hongyu Yu, Hua Chen
AISPO: Enhancing Depth Reliability for Robotic Manipulation of Non-Lambertian Objects via Affine-Invariant Shape Prior
IEEE Robotics and Automation Letters (**RA-L**), vol. 11, no. 7, pp. 7996–8003, 2026.

[2] Kaijun Wang, Zikai Ouyang, Xuping Wu, Jinyi Hong, Wei Pan, Haibo Lu, Jia Pan, Wei Zhang, **Linfang Zheng**
SSI-Policy: Learning Structured Scene Interfaces for Vision-Language Robotic Manipulation
IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.

[3] Fan Liu, Bernie Hao Zhu, **Linfang Zheng**, Wei Zhang
ESCAPO: Entropy-Guided Segment-Level Credit Assignment for Policy Optimization in VLA Post-Training
IROS 2026 Workshop on Building Scalable Infrastructure for Robot Learning.

[4] Chuye Zhang*, Xiaoxiong Zhang*, Wei Pan, **Linfang Zheng**†, Wei Zhang†
Generative Visual Foresight Meets Task-Agnostic Pose Estimation in Robotic Table-Top Manipulation
9th Conference on Robot Learning (**CoRL**), 2025.

[5] Tze Ho Elden Tse, Runyang Feng, **Linfang Zheng**, Jiho Park, Yixing Gao, Jihie Kim, Aleš Leonardis, Hyung Jin Chang
Collaborative Learning for 3D Hand-Object Reconstruction and Compositional Action Recognition from Egocentric RGB Videos Using Superquadrics
39th AAAI Conference on Artificial Intelligence (**AAAI**), 2025.

[6] **Linfang Zheng**, Tze Ho Elden Tse, Chen Wang, Yinghan Sun, Hua Chen, Aleš Leonardis, Wei Zhang, Hyung Jin Chang
GeoReF: Geometric Alignment Across Shape Variation for Category-Level Object Pose Refinement
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.

[7] Yinghan Sun, **Linfang Zheng**, Hua Chen, Wei Zhang
Multi-Resolution Planar Region Extraction for Uneven Terrains
IEEE International Conference on Robotics and Automation (**ICRA**), 2024.

[8] **Linfang Zheng**, Chen Wang, Yinghan Sun, Esha Dasgupta, Hua Chen, Aleš Leonardis, Wei Zhang, Hyung Jin Chang
HS-Pose: Hybrid Scope Feature Extraction for Category-Level Object Pose Estimation
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.

[9] Nora Horanyi, **Linfang Zheng**, Eunji Chong, Aleš Leonardis, Hyung Jin Chang
Where Are They Looking in the 3D Space?
IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (**CVPRW**), 2023. **Best Paper Award**.

[10] **Linfang Zheng**, Aleš Leonardis, Tze Ho Elden Tse, Nora Horanyi, Hua Chen, Wei Zhang, Hyung Jin Chang
TP-AE: Temporally Primed 6D Object Pose Tracking with Auto-Encoders
IEEE International Conference on Robotics and Automation (**ICRA**), 2022.

[11] Zhongqun Zhang, Wei Chen, **Linfang Zheng**, Aleš Leonardis, Hyung Jin Chang
Trans6D: Transformer-Based 6D Object Pose Estimation and Refinement
European Conference on Computer Vision Workshops (**ECCVW**), 2022.

[12] Hua Chen, **Linfang Zheng**, Wei Zhang
Optimal Control Inspired Q-Learning for Switched Linear Systems
American Control Conference (**ACC**), 2020.

*Equal contribution. †Corresponding author.

Manuscripts Under Review

[U1] **Linfang Zheng** (first author)
Journal Manuscript (Title Withheld During Review)
Major revision submitted to The International Journal of Robotics Research (**IJRR**).

[U2] **Linfang Zheng** (corresponding author)
Journal Manuscript (Title Withheld During Review)
Revision invited; resubmission in preparation for IEEE Robotics and Automation Letters (**RA-L**).

[U3] **Linfang Zheng** (corresponding author)
Conference Manuscript (Title and Venue Withheld During Double-Anonymous Review)
Under double-anonymous review.

[U4] **Linfang Zheng** (project supervisor and co-author)
Journal Manuscript (Title Withheld During Review)
Revision invited; resubmission in preparation for IEEE Transactions on Control Systems Technology (**TCST**).

[U5] **Linfang Zheng** (co-author)
Journal Manuscript (Title Withheld During Review)
Under review at IEEE Transactions on Robotics (**T-RO**).

[U6] **Linfang Zheng** (project supervisor and co-author)
Conference Manuscript (Title and Venue Withheld During Double-Blind Review)
Under double-blind review.

Preprints

[P1] Luqing Luo, Shichu Sun, Jiangang Yang, **Linfang Zheng**, Jinwei Du, Jian Liu
Object Gaussian for Monocular 6D Pose Estimation from Sparse Views
arXiv:2409.02581, 2024.

Honors & Awards

Best Paper Award, CVPR Workshop on GAZE, Vancouver, Canada	2023
Outstanding Master's Graduate, Harbin Institute of Technology	2017
Silver Award for Outstanding Master's Thesis, Harbin Institute of Technology	2017
First Prize Scholarship, Harbin Institute of Technology	2011 – 2017

Peer-Review Activity

Conferences	NeurIPS, CVPR, ECCV, ICCV, ICRA, IROS, CASE, AAAI, RSS, CoRL
Journals	IEEE Transactions on Robotics (T-RO), IEEE Robotics and Automation Letters (RA-L)

Skills

Programming	Python, C, MATLAB, Verilog; PyTorch, TensorFlow
Hardware	Embedded hardware and software design, integrated circuit design, FPGA
Languages	English, Mandarin (native)